DATA ANALYSIS PROJECT

# WWI
# COMPREHENSIVE SALES & PROFITABILITY ANALYSIS

An end-to-end analysis of sales performance, profitability drivers, customer behavior, and product trends to drive data-driven decisions.


### SALES OVERVIEW
Deep dive into overall sales performance and key metrics.

### PROFITABILITY ANALYSIS
Identify profit drivers and analyze margins across categories and segments.

### CUSTOMER INSIGHTS
Understand customer behavior, buying patterns, and segmentation for better targeting.

### PRODUCT PERFORMANCE
Analyze product trends, top performers, and underperformers to optimize product strategy.


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WWI
WORLD WIDE IMPORTERS

# WWI-Comprehensive-Sales-Profitability-Analysis

## Wide World Importers: Sales Performance & Financial Analysis

### Business Problem

The organization faced a critical challenge: raw, fragmented enterprise data that lacked actionable visibility. Specifically:

- **Lack of Performance Visibility**: Unable to distinguish between high-performing and underperforming sales regions, hindering strategic resource allocation.
- **Profitability Blind Spots**: No clear correlation between sales volume and net profit, leading to unidentified cost drivers and margin erosion.
- **Operational Inefficiency**: Ineffective analysis of inventory and buying packages, complicating supply chain and distribution management.

**Project Overview & Business Problem**

The organization faces a critical need to evaluate its historical sales performance and profitability across various dimensions, including geography, stock items, and sales channels. Despite generating significant revenue, there is a lack of visibility into:

- Identifying top-performing and underperforming sales regions.
- Understanding the correlation between sales volume, cost of goods sold (COGS), and net profit.
- Analyzing the distribution of product sales across different buying packages to optimize inventory and logistics.

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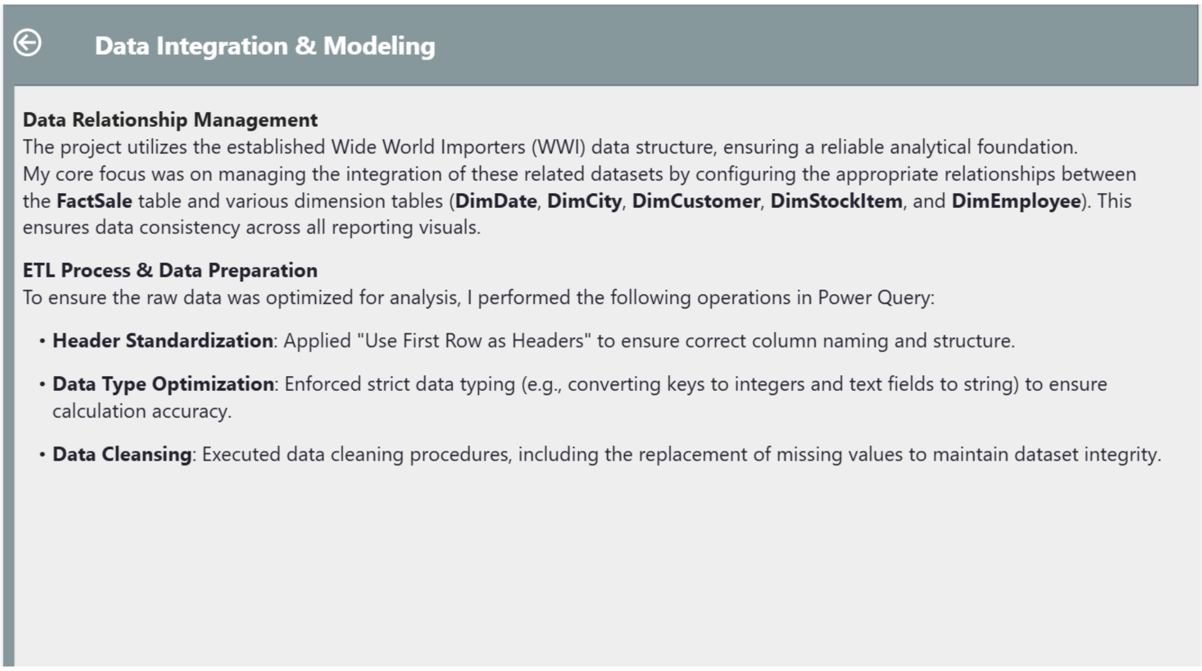
- **Dataset:** The data is sourced from the **Wide World Importers (WWI)** sample database, a standard enterprise-level dataset provided by Microsoft.
- **Data Structure:** The project utilizes a Star Schema architecture, featuring a central fact table (FactSale) connected to several dimension tables including:
    - DimDate: For temporal analysis.
    - DimCustomer & DimCity: For geographical and client-based segmentation.
    - DimStockItem: For granular product analysis.
    - DimEmployee: For sales performance tracking.

*Project_Overview & Business_Problem*

### The Solution

I engineered an end-to-end analytical solution to transform this data into a strategic asset:

- **Data Integration & Modelling**: Configured a robust Star Schema, establishing a "single source of truth" by linking the central `FactSale` table to key dimension tables (`DimDate`, `DimCity`, `DimCustomer`, `DimStockItem`, `DimEmployee`).
- **ETL Pipeline**: Built an automated Power Query pipeline to standardize headers, enforce strict data typing for calculation accuracy, and perform proactive data cleansing (handling missing values).



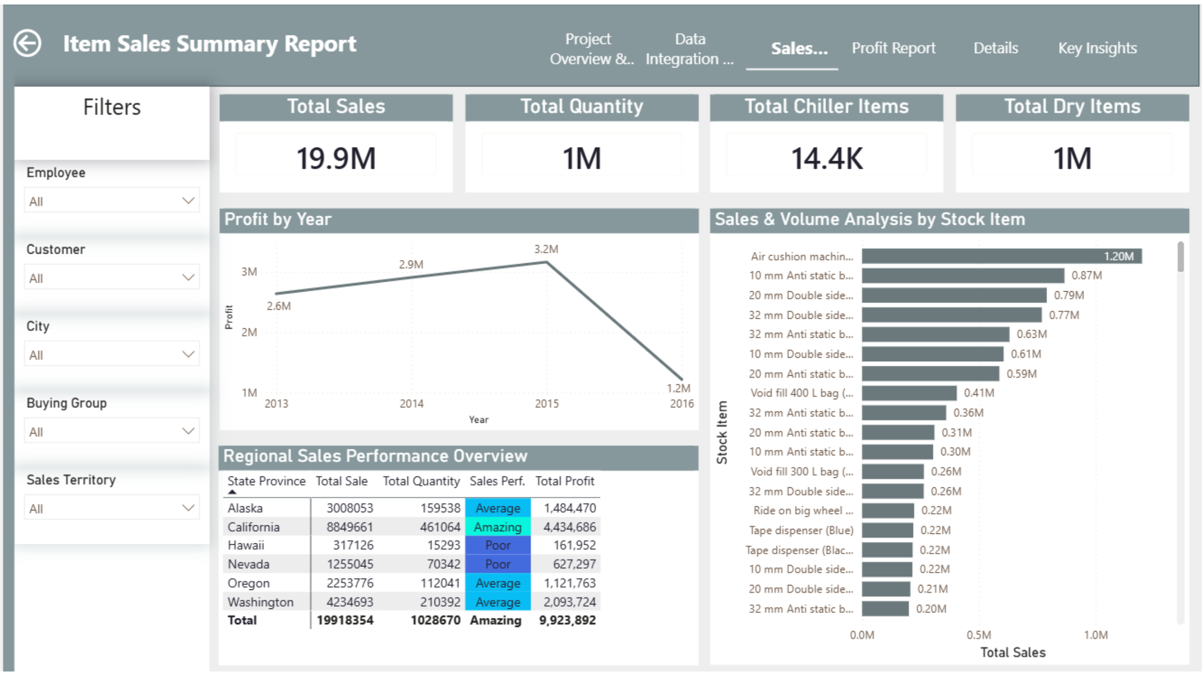

*Overview of the data modelling and the ETL transformation steps.*

## Analytical Dashboards

The solution comprises a three-tiered reporting system designed for deep-dive analysis:

### 1. Sales Performance Report

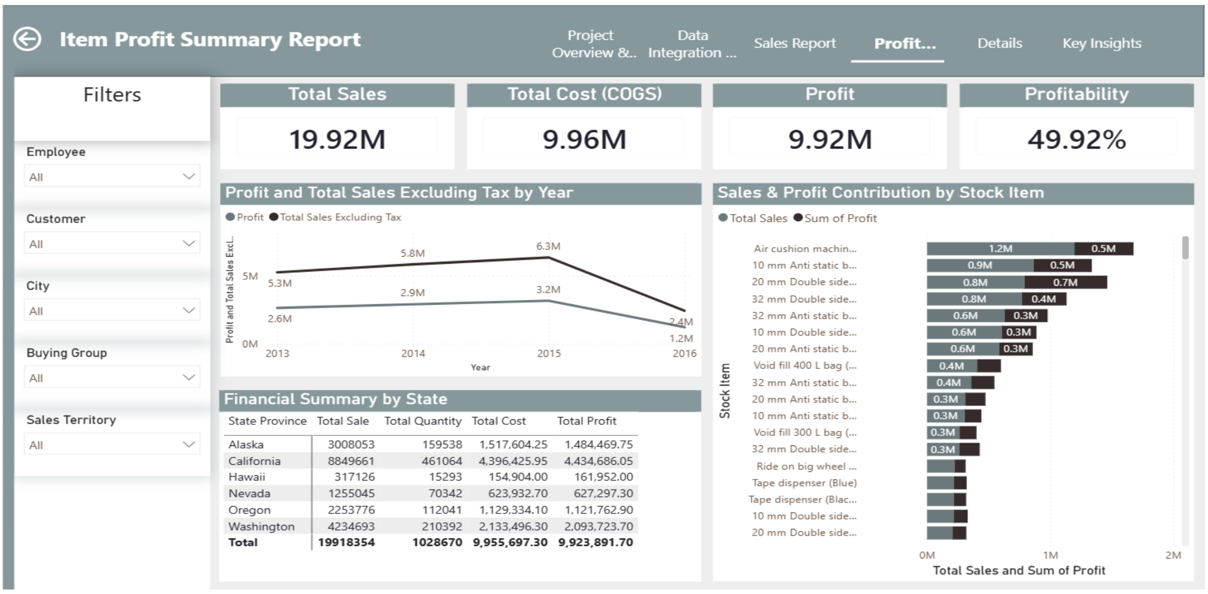
Focuses on top-line revenue, volume metrics, and regional performance benchmarks to guide sales strategy.




*Visualizing total sales, volume, and regional performance benchmarks.*

## 2. Profit & Financial Report

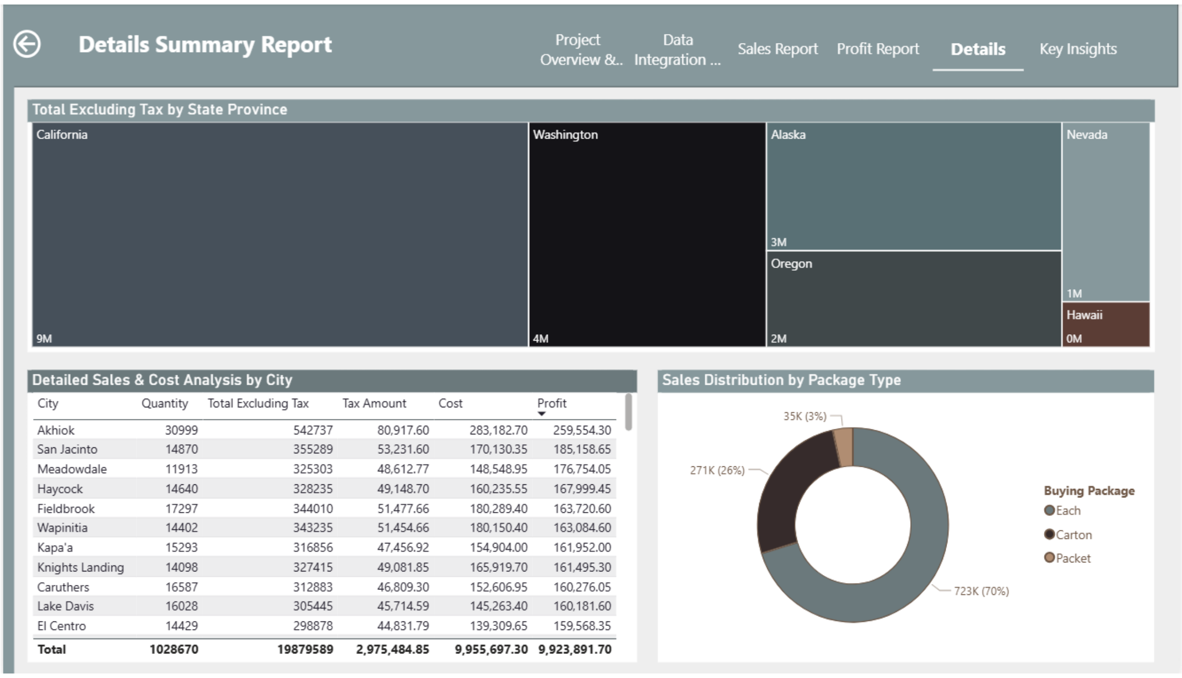
Bridges the gap between revenue and cost, highlighting profit margins and financial health per product line.




*Analysis of cost-to-profit correlations for individual stock items.*

### 3. Details & Distribution Analysis

Provides granular city-level analytics and examines the impact of buying packages on distribution efficiency.

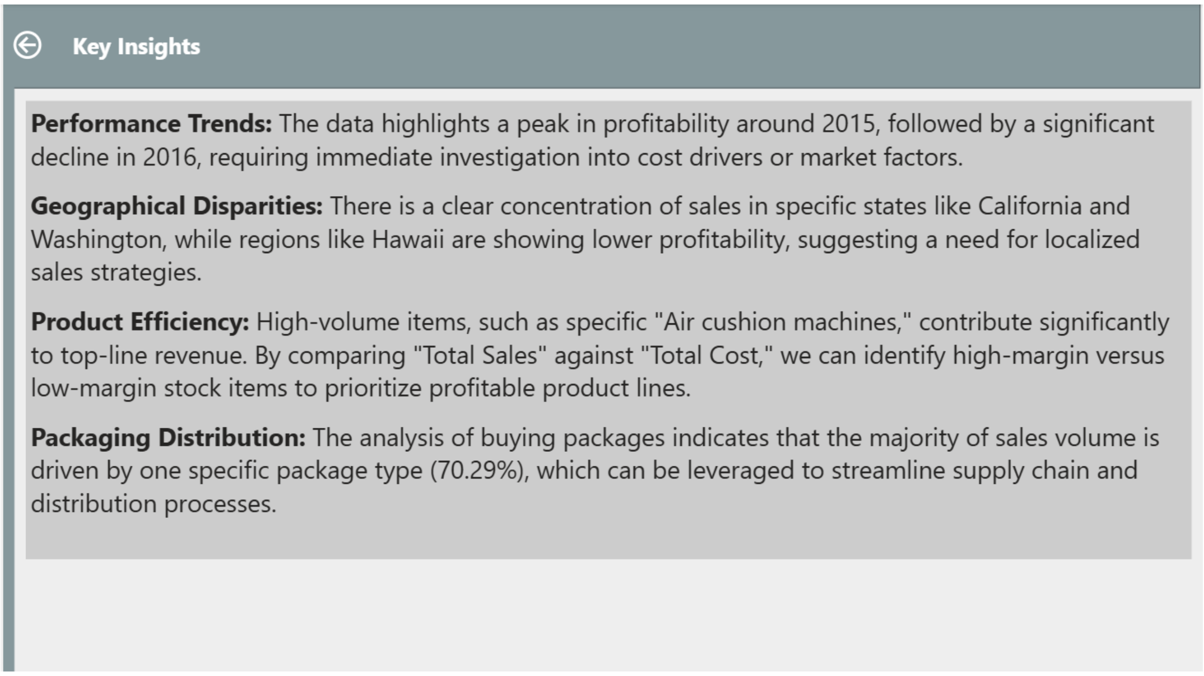



*Deep-dive analytics into city-level performance and supply chain trends.*

### Key Insights

The project revealed critical strategic findings that directly support data-driven decision-making:

- **Profitability Trends**: Identified a profit peak in 2015 followed by a decline in 2016, highlighting the need for immediate cost structure review.
- **Regional Disparities**: Highlighted California and Washington as primary revenue drivers, while identifying Hawaii as an area requiring localized sales intervention.
- **Product Efficiency**: Correlated sales volume with COGS to prioritize high-margin stock items.
- **Logistics Optimization**: Discovered that one specific buying package drives 70.29% of volume, providing a clear target to streamline distribution processes.




*Summary of key findings and actionable business recommendations.*

## Technical Skills Demonstrated

- **Data Engineering**: ETL pipeline design, data cleaning, and relationship management.
- **Business Intelligence**: Converting enterprise raw data into strategic, actionable dashboards.
- **Analytical Thinking**: Translating complex financial data into clear business insights.

GitHub Link: https://github.com/ebrahimqlini/WWI-Comprehensive-Sales-Profitability-Analysis/tree/main